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Course: Computer Vision

Course Code: ITA0506

Q1.Computer Vision Case Study

These questions are **analytical case studies**, so **code is not required**. The answers should focus on comparison, evaluation, justification, and practical deployment decisions.

1. Real-Time Detection System

Case

A surveillance system has three possible detection techniques:

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

The system operates under **strict latency constraints**, so both detection accuracy and processing speed must be considered.



Fig.1



Fig.2

Comparison

1. Viola-Jones

Viola-Jones provides **78% accuracy** and has **fast processing speed**. Its main advantage is low computational cost, making it suitable for systems with very limited processing resources.

However, its accuracy is considerably lower than YOLO and Deep Learning. In a surveillance environment, this may result in missed objects or incorrect detections.

2. YOLO

YOLO provides **92% accuracy** with **moderate speed**. It offers a strong balance between detection accuracy and processing time.

It is capable of detecting multiple objects in a single frame and is well suited to real-time computer vision applications.

3. Deep Learning

The Deep Learning method gives the highest accuracy of **95%**, but it is classified as **slow** in the given case.

Although it provides better detection performance, its higher computational requirement may violate the strict latency requirements of the surveillance system.

Evaluation:

The important trade-off is:

- Viola-Jones → fastest but lowest accuracy
- YOLO → high accuracy with moderate speed
- Deep Learning → highest accuracy but slowest



Fig.3

Recommended Method: YOLO

- **YOLO should be selected** because it provides **92% accuracy with moderate processing speed**.
 - The difference in accuracy between YOLO and Deep Learning is:
 - Thus, the system sacrifices only **3 percentage points of accuracy** compared with Deep Learning while gaining better suitability for real-time operation.

Deployment Justification:

For a surveillance system with strict latency constraints:

- Very fast detection is essential.
- Some accuracy loss can be accepted if it significantly improves response time.
- YOLO provides a practical balance between speed and accuracy.
- It is more accurate than Viola-Jones while being faster than the given Deep Learning method.

Final Decision:

YOLO is the most appropriate choice because it provides high accuracy while maintaining moderate speed, making it a practical compromise for real-time surveillance under latency constraints.

Q2. Robust Object Detection:

The performance of the three methods under different environmental conditions is:

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%



Fig.1



Fig.2

1. Performance under Normal Conditions:

Under normal conditions:

- Viola-Jones = **80%**
- YOLO = **90%**
- Deep Learning = **95%**
- Deep Learning provides the highest accuracy.
- The difference between Deep Learning and Viola-Jones is:
- Therefore, Deep Learning is considerably more accurate under normal conditions.

2. Performance under Occlusion:

When objects are partially hidden:

- Viola-Jones = **60%**
- YOLO = **75%**
- Deep Learning = **85%**

Deep Learning performs best.

Compared with Viola-Jones:

- Therefore, Deep Learning provides **25 percentage points higher accuracy** than Viola-Jones under occlusion.
- This indicates better robustness when parts of an object are hidden.

3. Performance under Low-Light Conditions:

Under low-light conditions:

- Viola-Jones = **55%**
- YOLO = **70%**
- Deep Learning = **80%**

Again, Deep Learning performs best.

Improvement over Viola-Jones:

- Therefore, Deep Learning is considerably more robust under poor lighting.

Overall Performance:

We can calculate the average performance.

- **Viola-Jones**
- **YOLO**
- **Deep Learning**

Comparison

Method	Average Accuracy
Viola-Jones	65%
YOLO	78.33%
Deep Learning	86.67%

Recommended Method: Deep Learning:

Deep Learning provides the **best overall performance**, with an average accuracy of **86.67%**.

Trade-offs

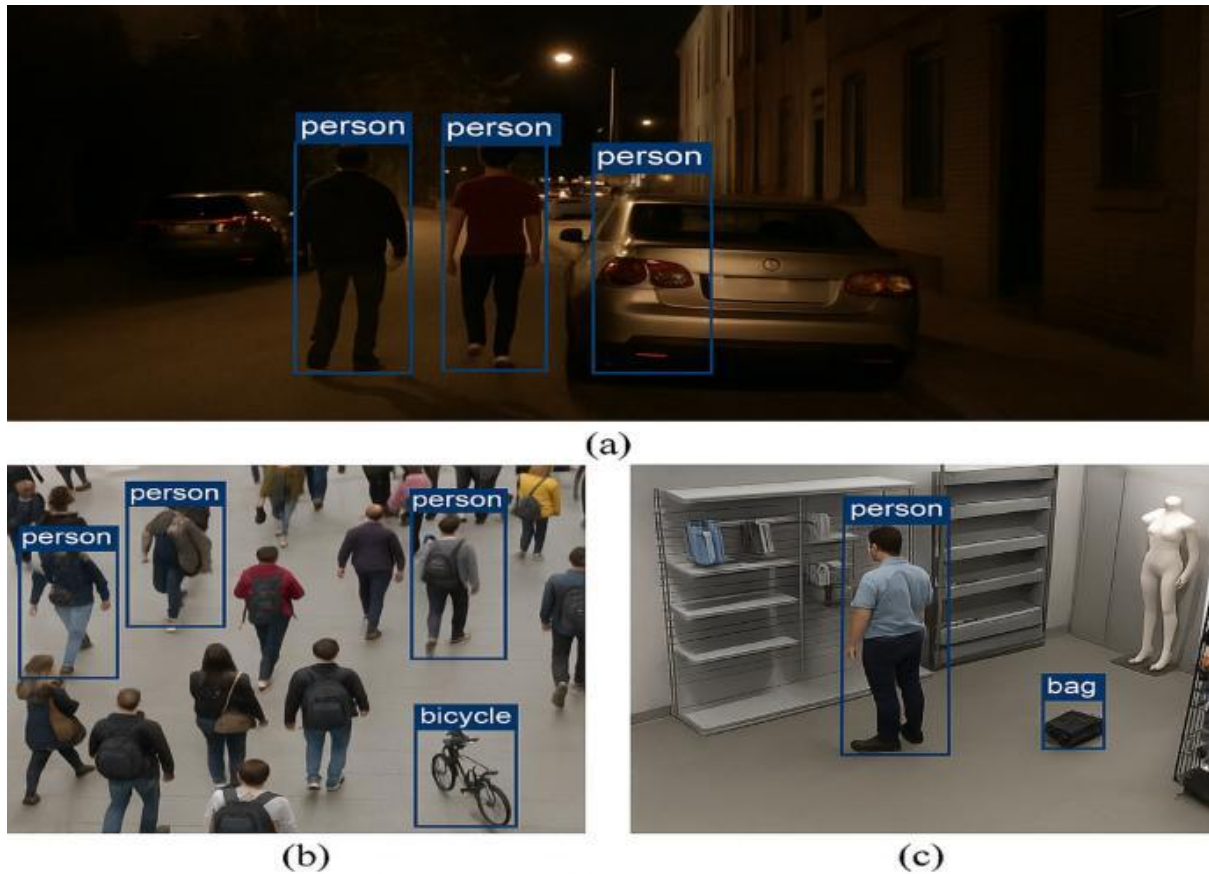


Fig.3

Advantages:

- Highest normal-condition accuracy
- Best performance with occlusion
- Best performance in low light
- Most robust overall

Disadvantages:

- Generally requires greater computational resources
- May require more processing time
- Deployment can be more expensive

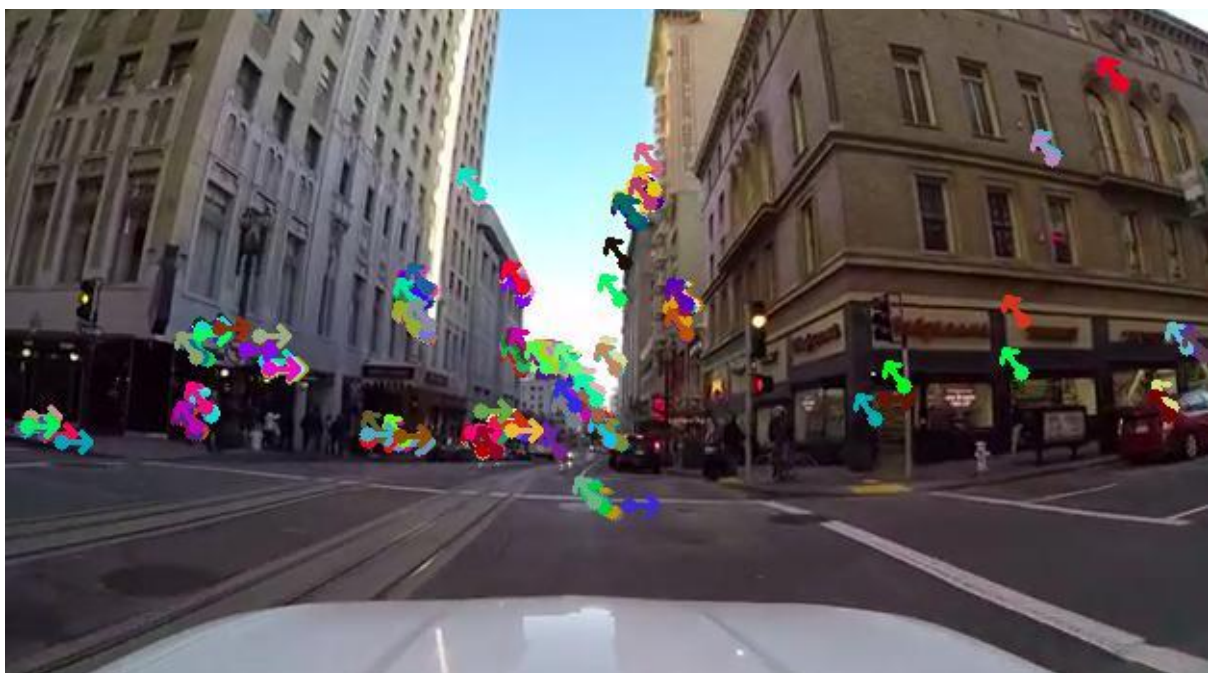
Final Decision:

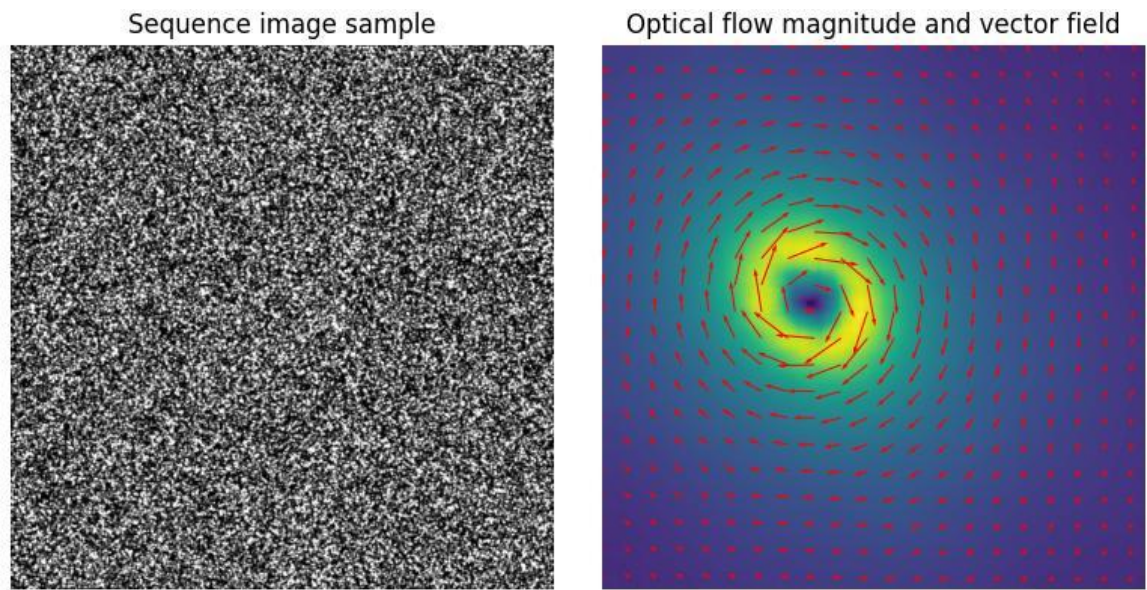
Deep Learning is the best method when robustness and detection accuracy are the primary requirements. YOLO is a suitable alternative when computational resources or real-time speed are more important. Viola-Jones is appropriate mainly when simplicity and low computational requirements are the priorities.

Q3.Optical Flow Analysis:

Two optical flow methods are available:

- **Method A:** Stable motion vectors but misses small motions.
- **Method B:** Detects fine motion but produces noisy vectors.





Case Interpretation

Optical flow estimates the apparent movement of pixels between consecutive frames. It is useful for:

- Motion tracking
- Vehicle tracking
- Human activity analysis
- Robot navigation
- Video surveillance

The two methods have different strengths.

Method A – Stable Motion Vectors

Method A produces **stable and consistent motion vectors**.

Advantages

- Low noise
- Consistent motion estimation
- Reliable tracking
- Suitable for long-term motion tracking

Limitation

Its major disadvantage is that it **misses small movements**.

For example, small movements of a person's hand or a small object may not be detected correctly.

Method B – Fine Motion Detection

Method B can detect **small and detailed movements**.

Advantages:

- Detects fine motion
- Sensitive to small changes
- Useful when small movements are important

Limitation:

It produces **noisy motion vectors**.

This can result in:

- Incorrect motion directions
- Unstable tracking
- False motion detection
- Difficulty in interpreting movement

Comparison

Factor	Method A	Method B
Motion stability	High	Low
Small-motion detection	Low	High
Noise sensitivity	Low	High
Tracking reliability	High	Moderate
Fine-motion analysis	Limited	Excellent



Evaluation:

For **dynamic scenes**, such as traffic surveillance or moving vehicles, stable motion estimation is generally important.

If the objective is to track the overall movement of:

- Vehicles
- People
- Large objects

then Method A is preferable because stable vectors provide reliable tracking.

- However, if the application specifically requires detection of **small and subtle movements**, Method B may be preferable.
- **Recommended Method: Method A for general dynamic-scene tracking**
- Method A is recommended when the main requirement is stable and reliable motion tracking.
- The reason is that noisy motion vectors can cause incorrect tracking even though Method B detects smaller movements.

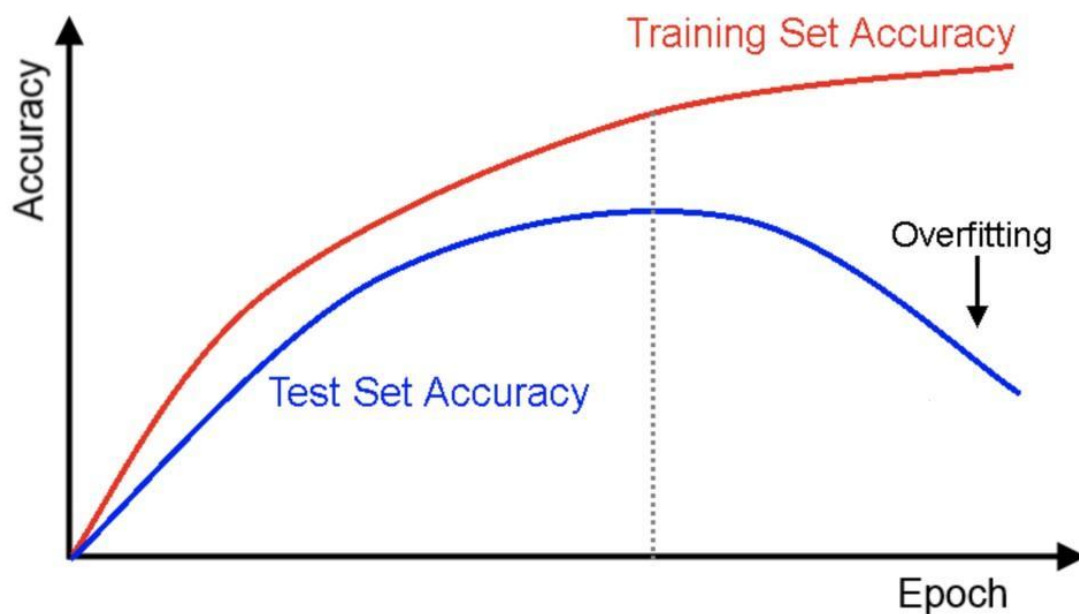
Final Decision:

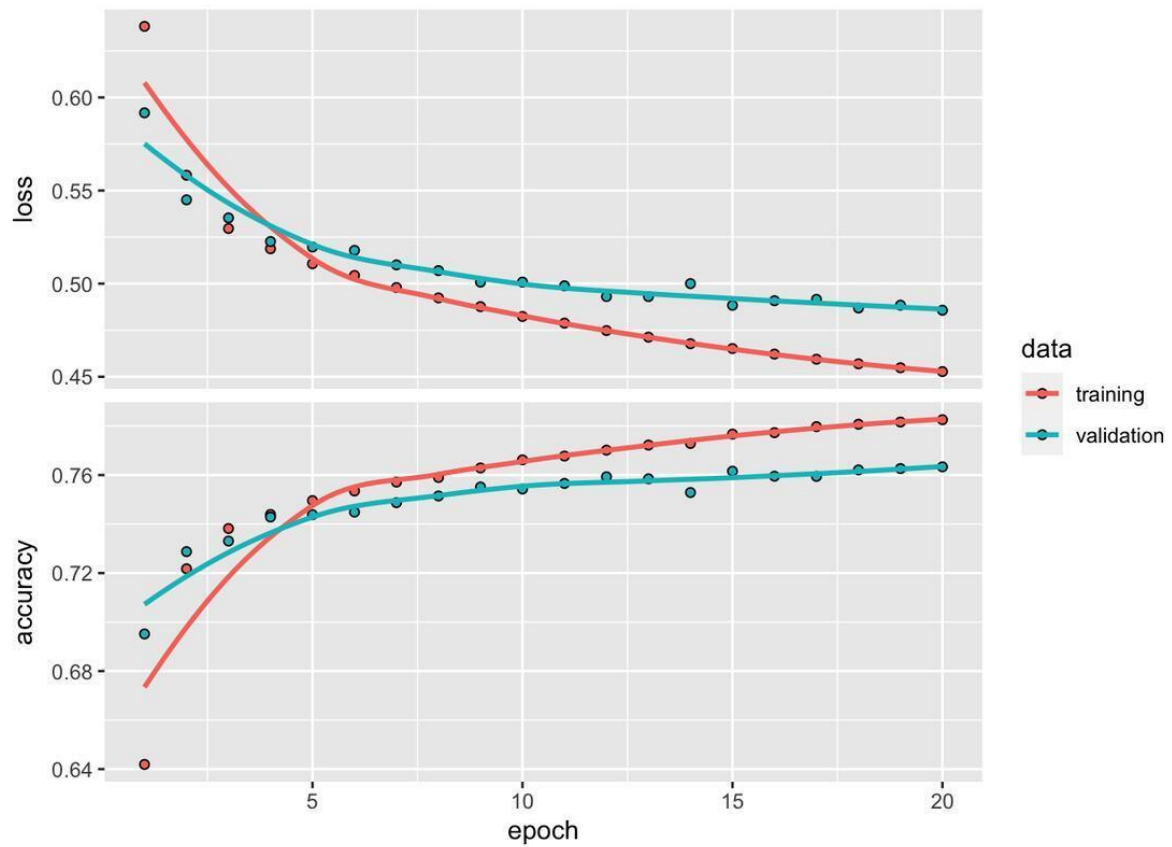
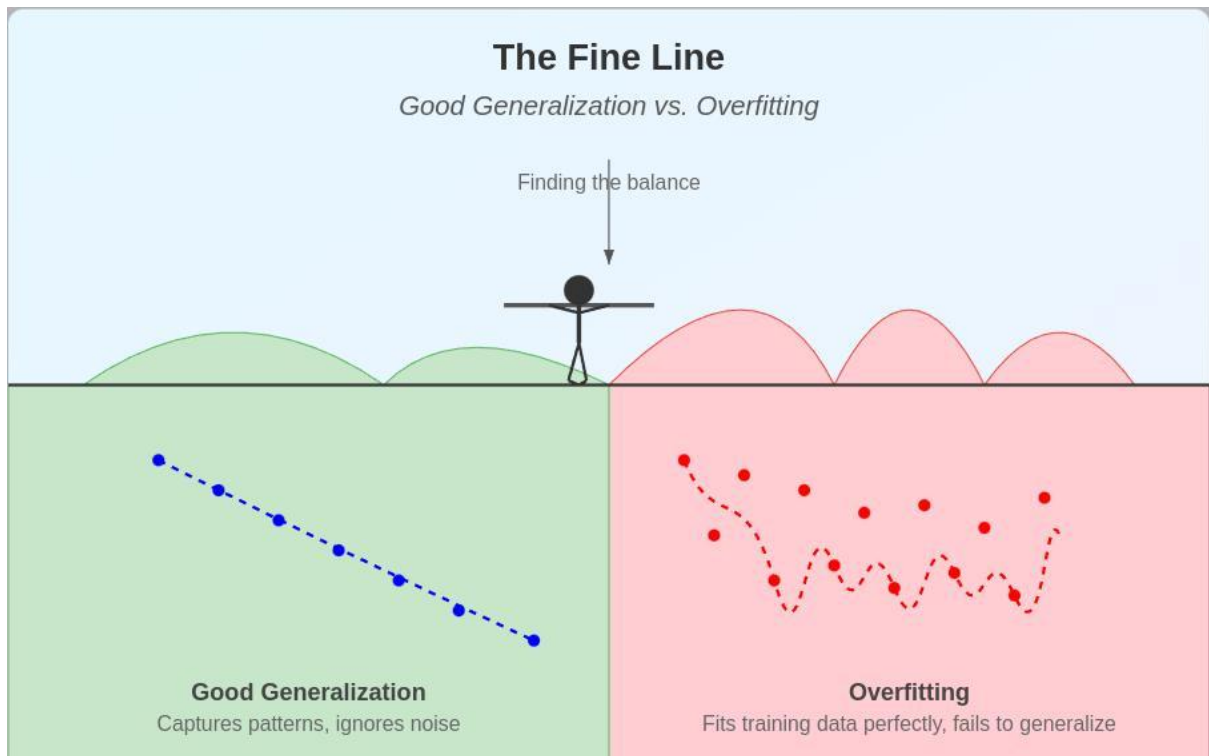
Method A is more suitable for general-purpose motion tracking in dynamic scenes because its stable motion vectors provide reliable tracking and are less affected by noise. Method B should be selected when detecting fine or small movements is more important than vector stability.

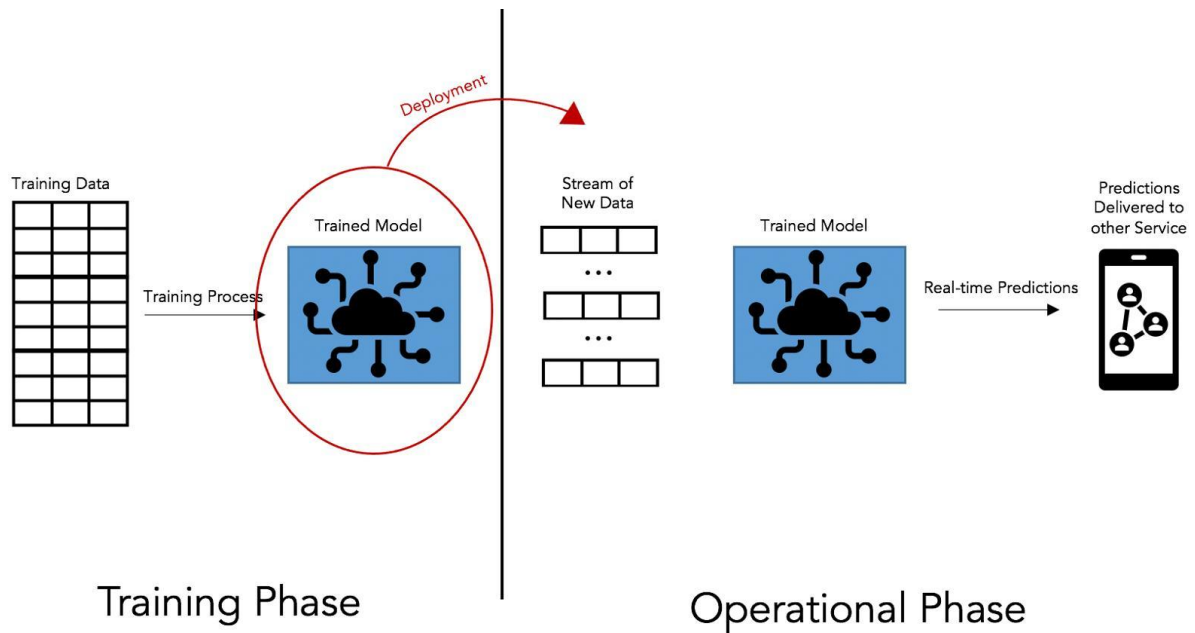
Q4. Model Generalization and Overfitting

The two models have the following performance:

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%







Model A Analysis:

There is a **28 percentage-point gap** between training and testing accuracy.

This indicates that the model performs extremely well on the training data but its performance decreases significantly on unseen data.

Interpretation:

Model A is likely suffering from **overfitting**.

Overfitting occurs when a model learns the training data too closely, including patterns that do not generalize well to new data.

Risk:

If deployed in the real world, Model A may produce unreliable predictions on new images or situations.

Model B Analysis:

The difference is:

The training-testing gap is only **2 percentage points**.

This indicates that Model B generalizes much better to unseen data.

Interpretation:

Model B has:

- Good training performance
- High testing performance
- Small generalization gap
- Lower overfitting risk

Comparison

Factor	Model A	Model B
Training accuracy	98%	90%
Testing accuracy	70%	88%
Accuracy gap	28%	2%
Overfitting risk	High	Low
Generalization	Poor	Good
Real-world reliability	Lower	Higher

Recommended Model: Model B

- Although Model A has a higher training accuracy, its testing accuracy is only 70%.
- Model B has a lower training accuracy but achieves 88% testing accuracy.
- Difference in testing accuracy:
- Therefore, Model B performs 18 percentage points better on testing data.

Final Decision:

Model B should be selected for real-world deployment because it has strong generalization ability, a small training-testing gap, high testing accuracy, and lower overfitting risk.

Q5. System Performance under Constraints

Three models are available:

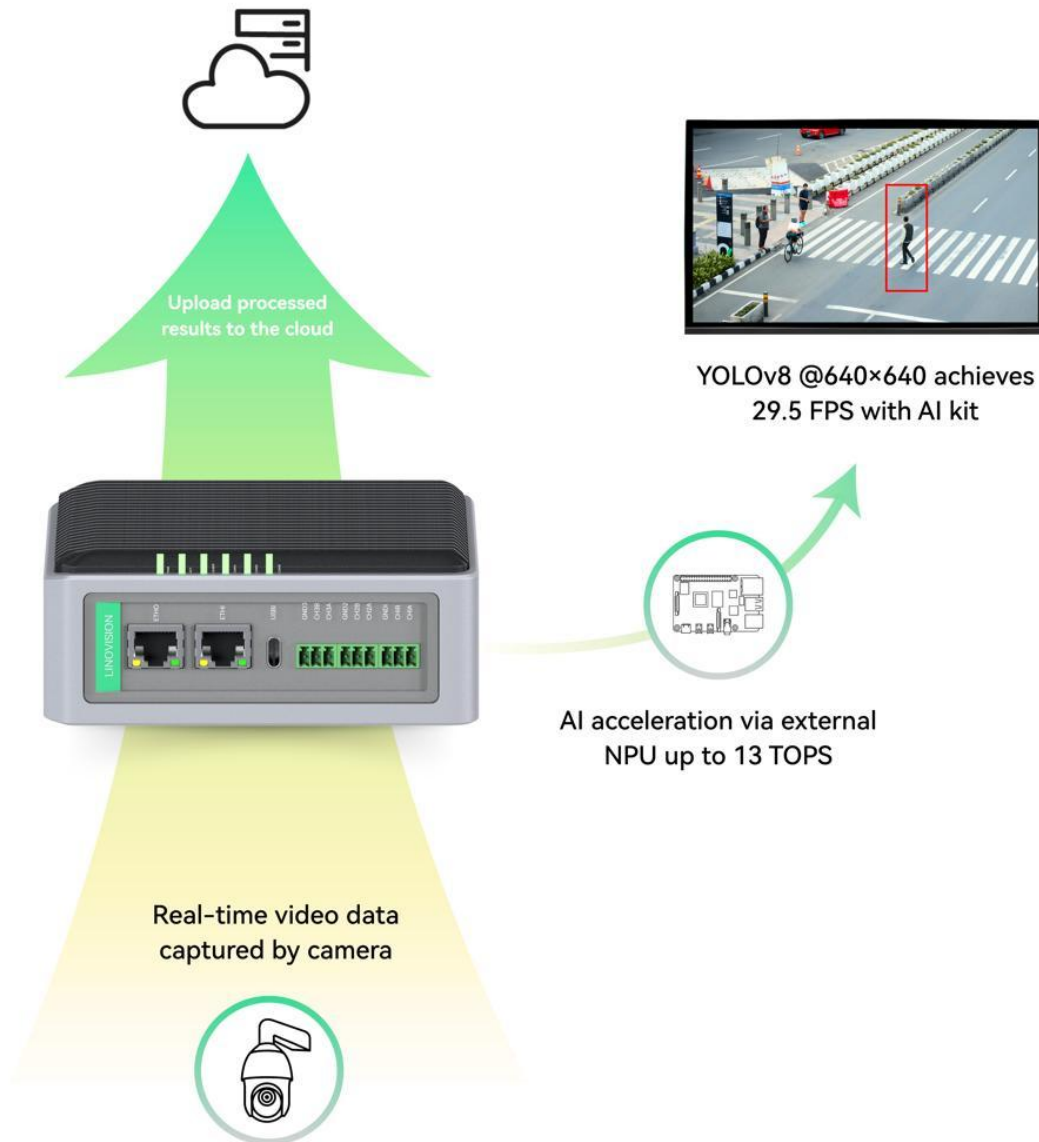
Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High







Edge Computing



Model A

- Accuracy = **88%**
- Latency = **50 ms**
- Cost = **Low**

Advantages

- Fastest model
- Lowest cost
- Suitable for strict real-time applications
- Requires fewer computational resources

Disadvantage:

Its accuracy is the lowest among the three models.

Model B

- Accuracy = **92%**
- Latency = **80 ms**
- Cost = **Moderate**

It provides a good compromise between accuracy, latency, and cost.

Compared with Model A:

So Model B improves accuracy by **4 percentage points**.

However, latency increases by:

Model C

- Accuracy = **95%**
- Latency = **120 ms**
- Cost = **High**

It provides the highest accuracy but also has the highest latency and cost.

Compared with Model B:

Model C provides only **3 percentage points additional accuracy**, but latency increases by:

and the cost also increases from moderate to high.

Overall Comparison

Factor	Model A	Model B	Model C
Accuracy	88%	92%	95%
Latency	50 ms	80 ms	120 ms
Cost	Low	Moderate	High
Speed	Excellent	Good	Low
Accuracy	Good	Very Good	Excellent
Overall balance	Good	Best	Lower

Recommended Model: Model B

For a real-time application, Model B provides the best overall balance.

Why not Model A?

Model A is faster and cheaper, but its accuracy is only **88%**.

If the application requires better detection reliability, the additional **4% accuracy** provided by Model B can justify the moderate increase in latency and cost.

Why not Model C?

Model C provides the highest accuracy, but:

- Latency is **120 ms**
- Cost is **High**
- It provides only **3% more accuracy than Model B**

Therefore, the additional accuracy may not justify the additional latency and deployment cost in a real-time system.

Practical Deployment Decision

The choice depends on application requirements:

If latency is extremely strict:

Model A

If balanced performance is required:

Model B

If accuracy is the highest priority and latency/cost are acceptable:

Model C

For the given **real-time application**, the best overall choice is:

Model B provides 92% accuracy, 80 ms latency, and moderate cost, making it the most practical balance between accuracy, response time, and deployment cost.